Aryaman Sharma

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Computational Data Science (MCDS) — May 2026 – May 2028

Netaji Subhas University of Technology (erstwhile NSIT) — Delhi, India
Bachelor of Technology | *Computer Engineering* | 8.70/10 CGPA — Aug. 2019 – May 2023

Work Experience

Machine Learning Engineer — September 2023 – Present
Fast Retailing (UNIQLO) — Tokyo, Japan

- Developed and deployed demand forecasting pipelines serving 4M+ SKUs across 40 business units, engineering temporal/categorical features that drove a $170M+ (¥26B) annual reduction in stockouts.
- Built JP-to-EN copy translation engine by fine-tuning frontier LLMs (via Azure OpenAI), designing custom guardrail pipelines (prohibited words, term mappings), and optimizing token throughput—reducing translation costs by $130K+ (¥20M) annually.
- Architected a multi-agent supply chain management system for production adjustment using OpenAI Agent SDK, engineered custom tool-calling schemas and parallel retrieval pipelines over 600MB+ relational databases, reducing decision cycle from 3 weeks to 2 days.
- Developed CockpitAI, a multi-agent report navigation system using 3 parallel analysis agents, dual RAG search, GPT-5 vision-based manual Q&A and cascading relevance scoring, optimizing multi-modal RAG search latency by 60% across corporate documents.

Research Associate (Collaborator) — Mar 2025–December 2025
University of South Carolina — Remote

- Co-developed CausalTrace (accepted AAAI 2026 IAAI), a neurosymbolic causal analysis agent within the SmartPilot industrial CoPilot — integrating causal discovery, counterfactual reasoning, and root cause analysis enriched by manufacturing ontologies and knowledge graphs, achieving MAP@3: 94%, PR@2: 97%, MRR: 0.92, and ROUGE-1: 0.91 agreement with domain experts
- Built Foresight, a time-series demand forecasting module for hourly yeast production, achieving 35% MSE improvement over LSTM baseline; and InfoGuide, an LLM-based conversational agent for operator Q&A over causal graphs and system documentation
- Designed evaluation framework based on the C3AN principles (Custom, Compact, Composite AI with Neurosymbolic Integration), scoring 4.59/5 across robustness, intelligence, and trustworthiness dimensions

Summer Research Intern — June 2022– August 2022
Vanderbilt University — Nashville, TN

- Achieved 96% accuracy in stop-level transit occupancy prediction using Transformer and LSTM architectures, optimizing resource allocation for paratransit systems (NSF-funded, Prof. Abhishek Dubey)

Selected Publications

CausalTrace: A Neurosymbolic Causal Analysis Agent for Smart Manufacturing
Accepted at *AAAI 2026: IAAI - Innovative Applications of AI Conference*

Designing Equitable Transit Networks
Accepted in the non-archival track at the *ACM Conference of EAAMO, 2023*

Concern Levels During COVID-19: An AI-Based Approach for Social Media Analysis
Published in *Springer's Innovations in Information and Communication Technologies, 2022*

Twitter Sentiment Analysis During Unlock Period of COVID-19 | ***R, RStudio***
Published in *IEEE's 2020 Sixth International Conference on Parallel, Distributed and Grid Computing (PDGC),2020*

Skills

ML& GenAI: PyTorch, TensorFlow, scikit-learn, Transformers, FAISS, Multi-Agent Systems (LangGraph, AutoGen), MCP, DSPy, RAG, RLHF/DPO
Systems & Engineering: Python, SQL, C++, FastAPI, Spark, Docker, Vertex AI, GCP, Azure OpenAI, PostgreSQL/pgvector, MongoDB, SQLite, NumPy, Git, pytest
Engineering: Python, SQL, C++, FastAPI, MongoDB, SQLite, pandas, NumPy, Git, pytest
Languages: English (Fluent), Hindi (Native), French (DELF B2), Japanese (JLPT N2)